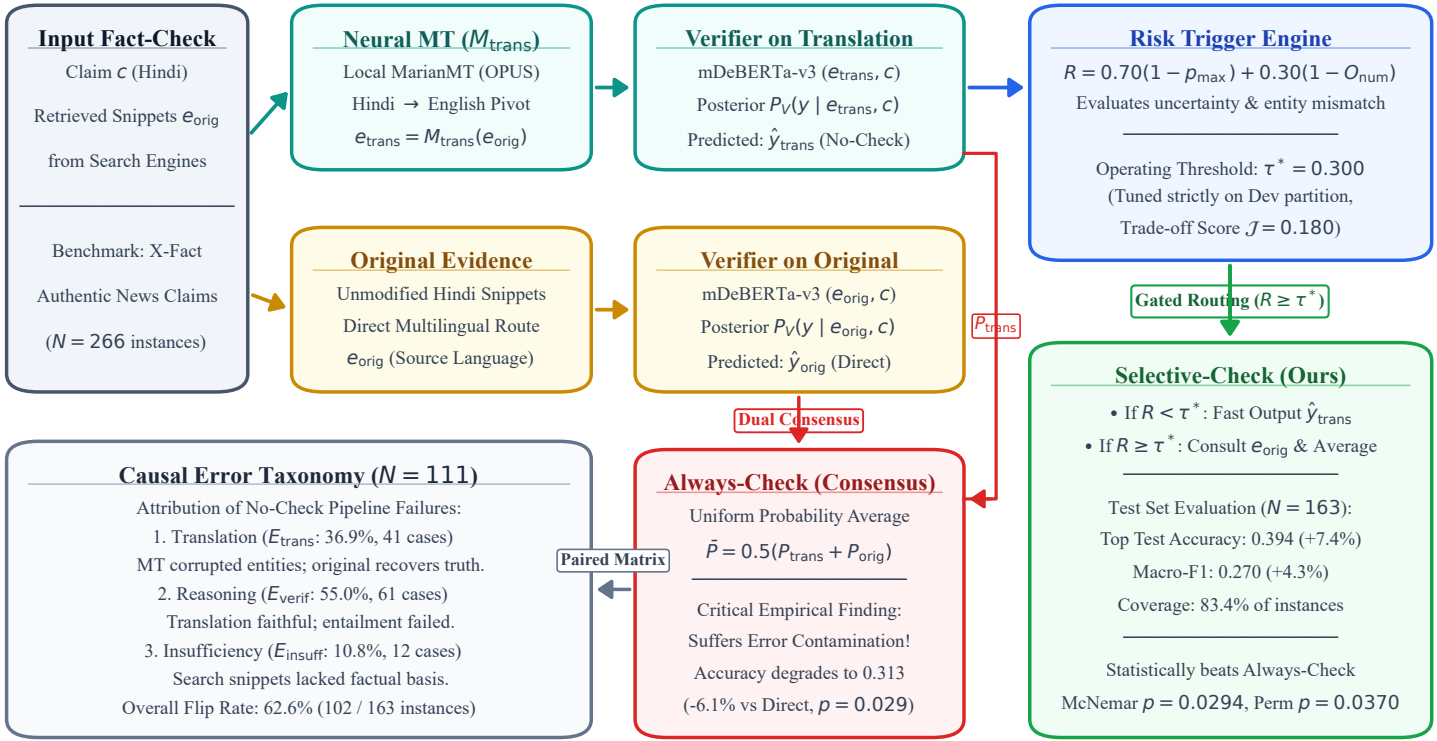

Input Fact-Check
Claim c (Hindi)
Retrieved Snippets e_{orig}
from Search Engines
Benchmark: X-Fact
Authentic News Claims
($N = 266$ instances)
Neural MT (M_{trans})
Local MarianMT (OPUS)
Hindi → English Pivot
$e_{trans} = M_{trans}(e_{orig})$
Original Evidence
Unmodified Hindi Snippets
Direct Multilingual Route
e_{orig} (Source Language)
Verifier on Translation
mDeBERTa-v3 (e_{trans}, c)
Posterior $P_V(y \mid e_{trans}, c)$
Predicted: $\hat{y}_{trans}$ (No-Check)
Verifier on Original
mDeBERTa-v3 (e_{orig}, c)
Posterior $P_V(y \mid e_{orig}, c)$
Predicted: $\hat{y}_{orig}$ (Direct)
P_{trans}
Dual Consensus
Always-Check (Consensus)
Uniform Probability Average
$\bar{P} = 0.5(P_{trans} + P_{orig})$
Critical Empirical Finding:
Suffers Error Contamination!
Accuracy degrades to 0.313
(-6.1% vs Direct, $p = 0.029$)
Paired Matrix
Causal Error Taxonomy ($N = 111$)
Attribution of No-Check Pipeline Failures:
1. Translation (E_{trans}: 36.9%, 41 cases)
MT corrupted entities; original recovers truth.
2. Reasoning (E_{verif}: 55.0%, 61 cases)
Translation faithful; entailment failed.
3. Insufficiency (E_{insuff}: 10.8%, 12 cases)
Search snippets lacked factual basis.
Overall Flip Rate: 62.6% (102 / 163 instances)
Risk Trigger Engine
$R = 0.70(1 - p_{max}) + 0.30(1 - O_{num})$
Evaluates uncertainty & entity mismatch
Operating Threshold: $\tau^* = 0.300$
(Tuned strictly on Dev partition,
Trade-off Score $\mathcal{J} = 0.180$)
Gated Routing ($R \geq \tau^*$)
Selective-Check (Ours)
• If $R < \tau^*$: Fast Output $\hat{y}_{trans}$
• If $R \geq \tau^*$: Consult e_{orig} & Average
Test Set Evaluation ($N = 163$):
Top Test Accuracy: 0.394 (+7.4%)
Macro-F1: 0.270 (+4.3%)
Coverage: 83.4% of instances
Statistically beats Always-Check
McNemar $p = 0.0294$, Perm $p = 0.0370$